Ashutosh Rai

ML/GenAI / Agentic AI Developer

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SUMMARY

IT professional with 9+ years of experience specializing in AI technologies and ML. GenAI Developer skilled in designing and building generative AI projects and agentic systems using the Lang graph framework, with backend development in Python and microservice architecture. Experienced in developing Retrieval-Augmented Generation (RAG) systems leveraging Langchain and vector databases such as PGVector and Croma db. Proven ability to deliver scalable, modular AI solutions optimized for real-time inference and adaptive learning.

EXPERIENCE

Senior Consultant

EY India

09/2021 - Present Pune, Bangalore, India

Software Engineer

Quale Infotech

01/2021 - 09/2021 Pune, India

Software Engineer

Webonise LAB

08/2018 - 01/2021 Pune, India

Associate

Paytm

08/2015 - 03/2017 Noida, India

SKILLS

EDUCATION

B.Tech (Elec & Comm)

AKTU

05/2011 - 05/2015 Greater Noida

Higher Secondary (Math)

Navals Hir. Sec. School

06/2007 - 06/2009 Gorakhpur (UP)

Secondary School

SSM Hir. Sec. School

06/2005 - 06/2007 Gorakhpur (UP)

CERTIFICATION

Python - PCEP

Blue Prims - AD01

Power Platform - PL-900 & PL-500

KEY ACHIEVEMENTS

- **Extraordinary Award – EY**

PROJECTS

I2I (Idea to Impact) – Agentic AI Procurement Platform (Allianz)

- Designed and implemented multi-agent AI workflows on Azure to automate enterprise task through tool calling and structured orchestration. Experienced in implementing LLM evaluation frameworks and matrices (e.g., relevance, groundness, correctness, latency, and cost metrics) using tools like LangSmith, RAGAS, and custom evaluation pipelines to continuously monitor and improve model performance
- Delivered measurable business impact: 40% improvement in project delivery speed, 60% boost in developer productivity, reduced onboarding time, and improved sprint planning and backlog management through contextual AI and enterprise tool integration.
- Input/Output validation guardrails using libraries like Guardrails AI (Rebuff/RAIL specs) and Pydantic schema enforcement for structured, safe responses
- Skills: Python, FastAPI, Azure OpenAI, Autogen, Redis, Langchain, RAG, PGVector, Hybrid Search, Git, AWS, Agile.

GenAI Lab – Enterprise Productivity Chatbot (Allianz)

- Developed a GenAI-driven chatbot platform with three key modules: a basic chat interface with document upload for enhanced responses, a marketplace featuring 100+ AI-powered tools (e.g., PPT generator, email writer, SharePoint copilot), and a public prompt library for sharing popular prompts.
- Increased personal productivity by 45%, enabling employees to streamline daily tasks and leverage GenAI Lab as a personalized AI assistant across the organization.
- Skills: Python, FastAPI, Azure OpenAI, Redis, LangChain, RAG, PGVector, Hybrid Search, Git, AWS.

Lydia – GenAI-Enriched Chatbot for Financial, Risk & Insurance Domains (HSBC)

- Developed Lydia, a GenAI-powered chatbot aimed at improving productivity and accuracy for risk, finance, and insurance teams.
- Implemented document upload and Retrieval-Augmented Generation (RAG) capabilities to enable users to query internal HSBC data for tasks like banking account opening and operational decision-making.
- Delivered business value by reducing manual errors, automating document generation, and improving access to organizational data and team-specific insights across multiple business functions.
- Skills: Microservice architecture, Python, FastAPI, Azure OpenAI, Azure, RAG, Pinecone, PostgreSQL, LangChain.

Emerald – Intelligent Automation

- Automated operational processes for HSBC's core banking system migration in the Philippines and Indonesia, including account, time deposit (TD), demand deposit (DD), loan, balance, and customer information transfers. Generated unique customer IDs through RPA-driven workflows.
- Delivered 20 FTE savings by automating data migration with 99% accuracy and resolving the remaining 1% through reconciliation reports and business-approved exceptions.
- Skills: RPA (Blue Prism), SharePoint, Excel, ServiceNow, Confluence, HSBC Account Opening Portal, Mainframe Applications, UI Automation.

ADDITIONAL RESPONSIBILITIES

- Led internal teams deployed onsite to ensure successful project delivery, consistently aligning with client expectations and delivery schedules.
- Directed the development of GenAI-focused Proof of Concepts (POCs) with interns to address client use cases and validate solutions.
- Presented product demonstrations to clients and senior leadership, effectively communicating technical capabilities and business value.
- Supported solution design, architecture, and roadmap development, incorporated stakeholder feedback, and played a key role i